

HUNGE, Sweden

WMO: 023310

Lat: 62.7504N

Lon: 15.0845E

Elev: 343

StdP: 97.27

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.8	(c) -24.3	(d) -30.8	(e) 0.2	(f) -27.7	(g) -26.9	(h) 0.3	(i) -24.3	(j) 9.1	(k) -1.1	(l) 8.1	(m) -1.3	(n) 0.2	(o) 0	(p) 0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.2	(c) 24.8	(d) 16.3	(e) 22.9	(f) 15.6	(g) 21.0	(h) 14.8	(i) 17.8	(j) 22.5	(k) 16.7	(l) 21.1	(m) 15.7	(n) 19.5	(o) 2.7	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.9	19.5	15.0	11.1	18.4	14.0	10.4	17.5	51.7	22.4	48.3	21.1	45.1	19.5	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.7	6.7	5.9	-30.6	27.0	3.6	2.0	-33.2	28.5	-35.3	29.7	-37.3	30.8	-39.9	32.3
			-30.9	18.8	3.3	1.6	-33.3	20.0	-35.3	20.9	-37.1	21.9	-39.5	23.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-8.3	-7.5	-4.0	1.8	7.3	11.6	14.5	12.9	8.3	2.5	-2.6	-6.6
	DBStd	9.27	6.88	6.63	5.59	3.46	3.69	3.25	3.06	2.94	3.18	3.61	4.85	6.88
	HDD10.0	3059	566	491	434	247	101	18	2	7	68	232	377	514
	HDD18.3	5767	824	724	693	497	342	201	125	169	302	489	627	773
	CDD10.0	341	0	0	0	0	17	68	141	98	17	1	0	0
	CDD18.3	8	0	0	0	0	0	1	6	1	0	0	0	0
	CDH23.3	128	0	0	0	0	3	25	83	17	0	0	0	0
	CDH26.7	10	0	0	0	0	0	1	9	1	0	0	0	0
(14) Wind	WSAvg	2.5	2.3	2.4	2.6	2.5	2.7	2.7	2.4	2.3	2.5	2.6	2.4	2.2
(15) Precipitation	PrecAvg	614	36	26	27	35	46	74	94	83	56	54	45	39
	PrecMax	839	66	53	58	107	78	130	198	137	116	139	98	69
	PrecMin	456	5	2	5	3	19	20	21	32	17	21	16	8
	PrecStd	96	18	13	12	23	20	29	49	30	28	30	20	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	6.2	8.9	17.0	23.6	25.8	27.8	25.5	19.2	13.9	8.9	6.0
		MCWB	3.1	2.9	4.6	9.0	13.5	15.3	18.1	17.2	13.8	11.0	7.6	4.1
	2%	DB	3.0	4.5	6.6	13.4	20.8	23.6	25.8	22.9	17.2	11.4	6.5	4.5
		MCWB	1.7	2.2	3.4	7.3	12.2	14.7	17.4	16.7	13.1	9.2	5.5	3.0
	5%	DB	1.7	2.9	4.9	10.8	17.9	21.2	23.5	20.9	15.5	9.8	4.7	3.2
		MCWB	0.7	1.4	2.3	5.9	11.0	14.1	16.4	15.5	12.2	8.0	3.9	2.0
	10%	DB	0.5	1.3	3.6	8.4	15.0	18.6	21.6	19.0	13.9	8.6	3.4	1.7
		MCWB	-0.4	0.1	1.4	4.4	9.6	12.7	15.6	15.0	11.4	7.1	2.6	0.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.4	5.1	9.8	14.8	17.2	19.7	18.7	15.1	11.3	7.6	4.5
		MCDB	4.5	5.4	8.3	15.7	21.1	23.1	25.2	23.1	18.0	13.0	8.7	5.6
	2%	WB	2.0	2.4	3.6	7.8	13.1	15.7	18.3	17.6	13.7	9.6	5.7	3.2
		MCDB	3.0	4.3	6.2	12.6	18.7	20.9	23.5	21.3	16.2	10.8	6.4	4.4
	5%	WB	0.8	1.4	2.5	6.2	11.7	14.6	17.2	16.5	12.7	8.5	3.9	2.0
		MCDB	1.5	2.8	4.7	10.0	16.9	19.8	22.1	19.7	14.9	9.6	4.5	3.0
	10%	WB	-0.4	0.2	1.5	4.8	10.2	13.4	16.2	15.4	11.7	7.2	2.7	0.8
		MCDB	0.4	1.3	3.4	8.0	14.7	18.1	20.7	18.3	13.6	8.5	3.4	1.5
(35) Mean Daily Temperature Range	MDBR		8.2	8.9	10.2	10.2	10.8	10.1	10.2	9.5	8.6	6.9	5.8	7.5
	5% DB	MCDBR	7.6	7.7	9.6	15.2	16.8	14.8	13.9	13.0	11.4	9.5	6.5	7.2
		MCWBR	6.9	6.1	7.0	9.4	8.9	7.1	6.7	7.2	7.5	7.1	5.7	6.3
	5% WB	MCDBR	7.2	7.2	9.2	13.5	14.5	12.8	12.4	11.4	10.1	8.5	6.2	7.1
		MCWBR	6.7	5.8	6.8	8.7	8.2	6.6	6.4	6.6	7.0	6.7	5.6	6.3
(40) Clear-Sky Solar Irradiance	taub		0.212	0.270	0.297	0.331	0.338	0.333	0.348	0.339	0.308	0.293	0.225	0.344
	taud		2.080	2.258	2.331	2.411	2.465	2.505	2.477	2.496	2.566	2.486	2.161	2.759
	Ebn at Noon		460	690	814	858	881	891	865	843	810	674	438	294
	Edh at Noon		31	59	80	93	95	94	94	85	65	48	28	19
(44) All-Sky Solar Radiation	RadAvg		0.20	0.81	2.23	3.84	4.64	5.32	4.86	3.69	2.28	1.00	0.28	0.10
	RadStd		0.02	0.11	0.22	0.31	0.47	0.41	0.57	0.32	0.26	0.16	0.05	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page